Pneumatics and Air Types

Turn this handle and the bus door opens by hand.



National 5 Engineering Science

Daily Review



1	**AND** gate, A = 1 and B = 0. **State** Z.	0
2	**State** another name for a **NOT** gate.	Inverter
3	A **push button**. **State** analogue or digital.	Digital
4	**State** the **unit** of **force**.	Newton (N)
5	**State** the formula for **work done**.	$E_w = Fd$

By the end of this lesson… (p.3)



1. I can **describe** what pneumatics is and **give** uses of it.
2. I can **name** the three air types: main air, pilot air and exhaust.
3. I can **describe** the function of main air and of pilot air.
4. I can **read** the pipes on a circuit: main air, pilot air, joined and crossing.
5. I can **follow** the safety rules for the pneumatics kit.

What has this handle done to the air in the door?

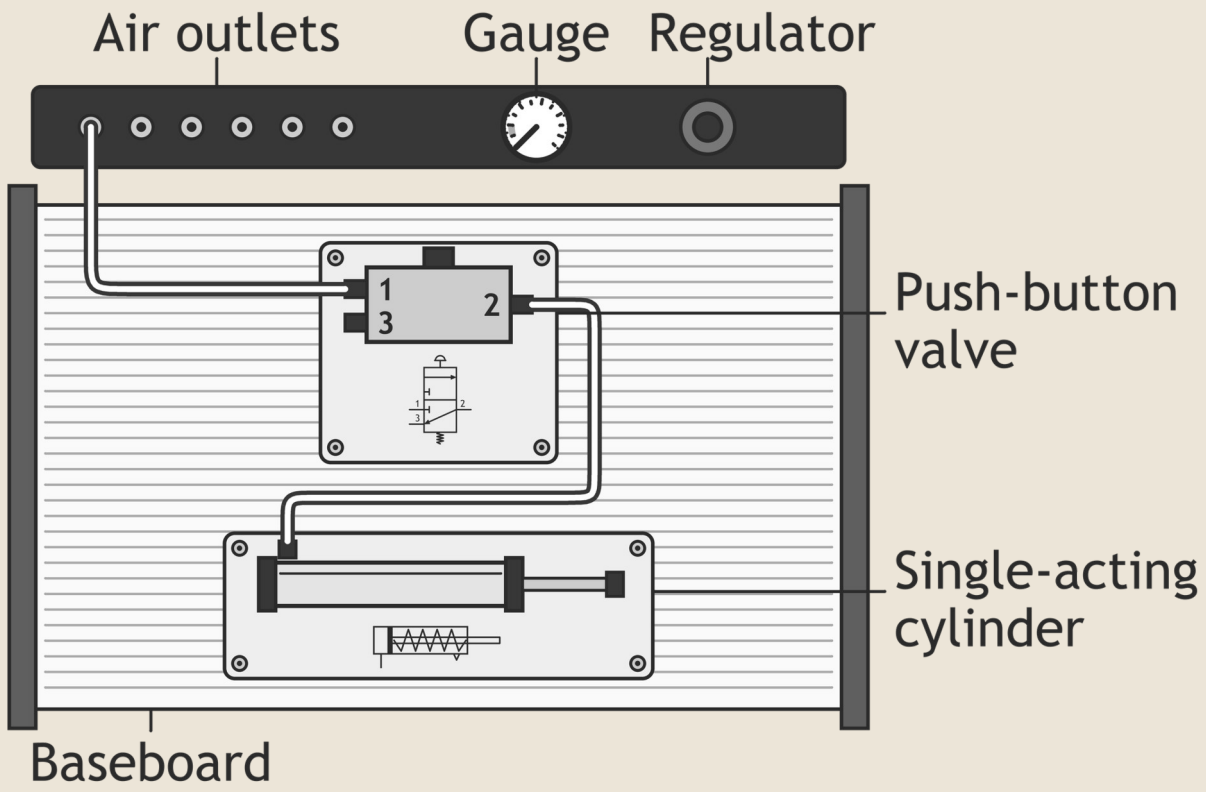


The Kit

The **regulator** sets the air **pressure**.
The **gauge** shows it.

The **valve** controls the air.

The **cylinder** moves.



Show Me: What Did You See?

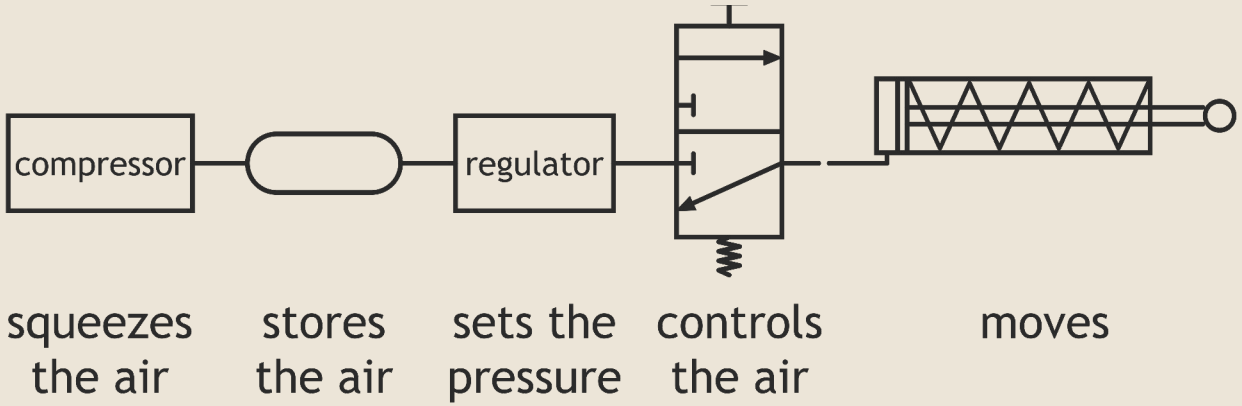
1. The button is **pressed**. **State** what the piston does.
2. The button is **released**. **State** what the piston does.
3. **State** what the **hiss** is.

Moves **out**

Moves back **in**

Used air **leaving**

What Is Pneumatics? (p.3)



Pneumatics uses **compressed** air to make things move.

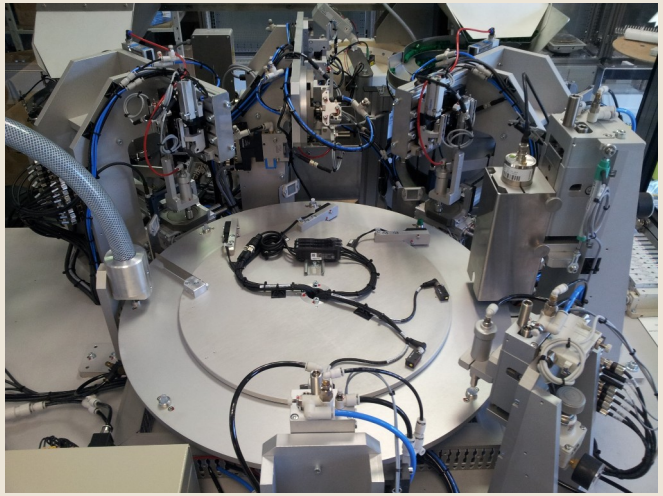
A **compressor** squeezes air and stores it in a **reservoir**.

A regulator sets the working **pressure** before the air reaches the circuit.

Where Pneumatics Is Used (p.3)

Pneumatics is used on **production** lines to push, clamp, lift and stamp products.

Bus and train doors and fairground rides also use **pneumatic** cylinders.





Air Types (p.3)

Symbol	Name	What it does
	**main air**	Main air operates a **cylinder**: it makes the piston outstroke or instroke.
	**pilot air**	Pilot air is a small air signal used to **actuate** a valve.
	**exhaust**	Exhaust is where used air leaves the circuit to the **atmosphere**.

Pipes on a Circuit (p.4)

main air pipe
A **solid** line carries main air.

pilot air pipe
A **dashed** line carries pilot air.



joined
A **dot** shows that the pipes are joined.



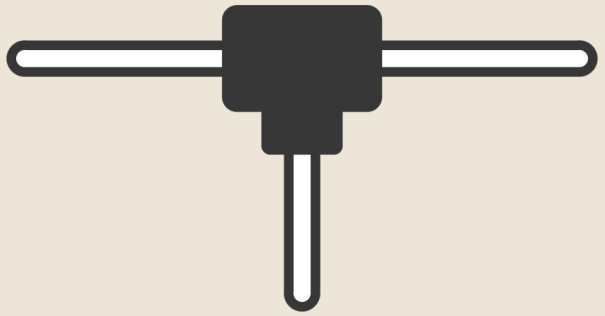
crossing
Pipes that cross without a dot are **separate** pipes.

Kit and Symbol

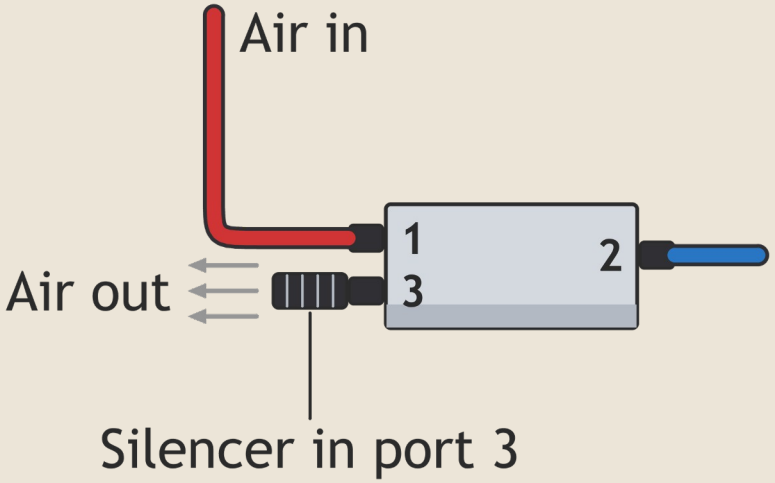
On the kit every pipe is a **blue tube**.

A **T-piece** joins pipes: a **dot** on paper.

The **silencer** in port 3 is the **exhaust**.









Show Me: Name the Symbol

1



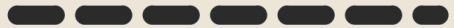
Main air

2



Exhaust

3



Pilot air pipe

4



Joined

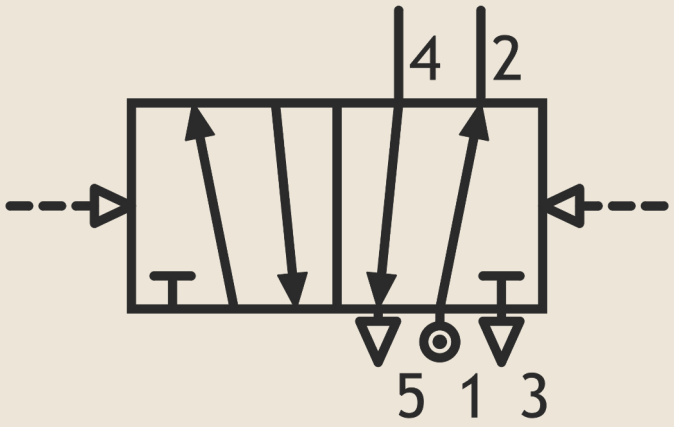
5



Crossing

Watch Out: Name Every Air Type

ON THE PAPER'S VALVE



2026: some candidates **did not name** these symbols.

THE CORRECT ANSWER



main air



exhaust



pilot air: dashed

Safety with the Kit (p.4)

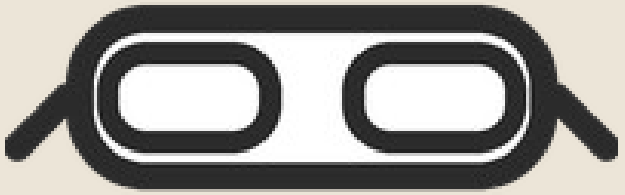
Wear **goggles** whenever the air is on.

Turn the air **off** before you connect or take apart any pipe.

Push each pipe fully **in** and pull gently to check it is held.

Never point a pipe or an exhaust at **anyone**.

Keep fingers clear of a **moving** piston.

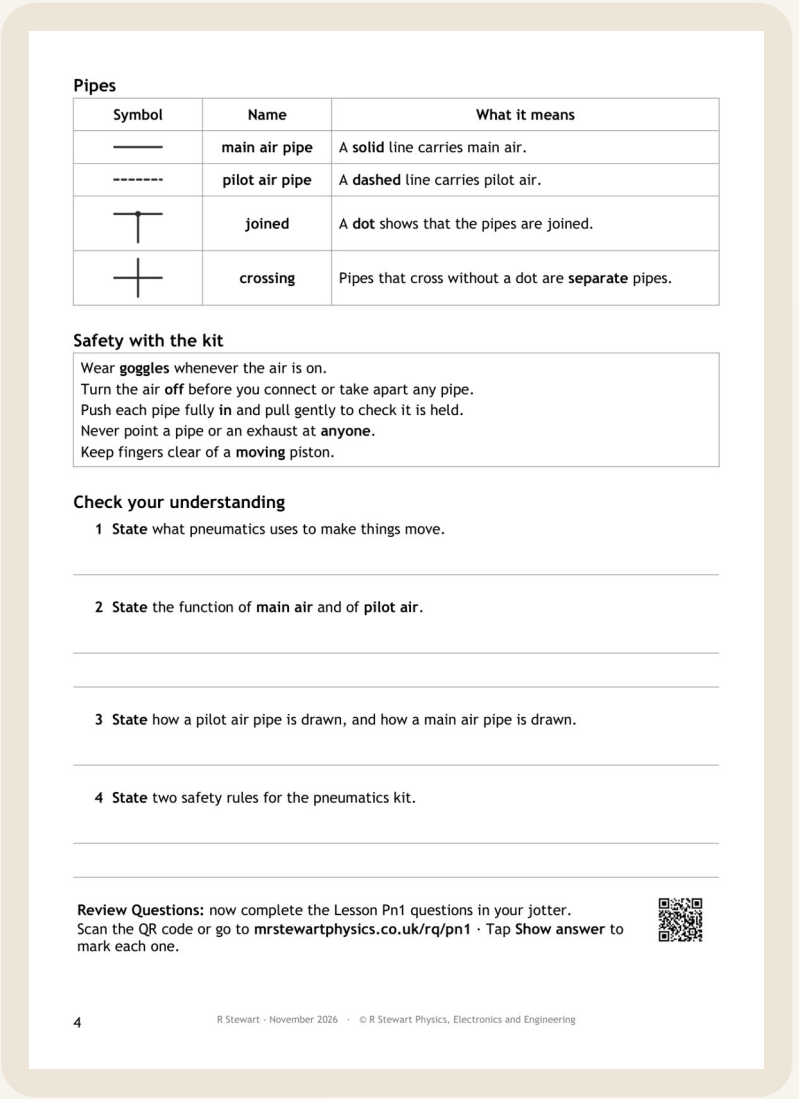


Check Your Understanding (pp 3-4)

In your booklet

Answer the **Quick check** on page **3**.

Answer Questions **1** to **4** on page **4**.



Pipes

Symbol	Name	What it means
———	**main air pipe**	A **solid** line carries main air.
- - - - - - -	**pilot air pipe**	A **dashed** line carries pilot air.
	joined	A **dot** shows that the pipes are joined.
	crossing	Pipes that cross without a dot are **separate** pipes.

Safety with the kit

Wear **goggles** whenever the air is on.
Turn the air **off** before you connect or take apart any pipe.
Push each pipe fully **in** and pull gently to check it is held.
Never point a pipe or an exhaust at **anyone**.
Keep fingers clear of a **moving** piston.

Check your understanding

1 **State** what pneumatics uses to make things move.

2 **State** the function of **main air** and of **pilot air**.

3 **State** how a pilot air pipe is drawn, and how a main air pipe is drawn.

4 **State** two safety rules for the pneumatics kit.

Review Questions: now complete the Lesson Pn1 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn1** · Tap **Show answer** to mark each one.

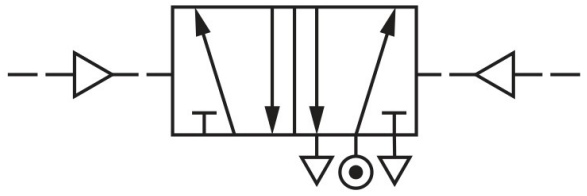
R Stewart · November 2026 · © R Stewart Physics, Electronics and Engineering

Answers: Check Your Understanding (pp 3-4)

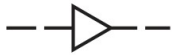
Question	Answer
Quick check	Through the **exhaust**, to the **atmosphere**.
1	**Compressed** air.
2	Main: operates a **cylinder**. Pilot: **actuates** a valve.
3	Pilot: a **dashed** line. Main: a **solid** line.
4	Any **two** of the **safety** rules on page **4**.

Exam: 2026 Section 1 Question 1

1. The symbol for a 5/2 valve is shown.



Complete the table below by naming the air type symbols shown on the valve. **3**

Air type symbol	Name of air type
	
	
	

3 marks

Name the **three** air types.

Boards up when you have all three.

Marking: 2026 Section 1 Question 1

Question	Expected response	Max mark	Additional guidance
1	main exhaust pilot	3	1 mark for main air. 1 mark for exhaust. 1 mark for pilot air. Accept air types clearly labelled on the valve.

Checkpoint

1. **Explain** why a pilot air pipe is drawn **dashed**.

   A **signal**, **so** drawn unlike main air

2. **State** what a **regulator** sets.

   The working **pressure**

3. **State** where **exhaust** air goes.

   To the **atmosphere**

Review Questions (p.4)

Lesson Pn1 (online)

Answer the Lesson **Pn1** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn1**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this lesson… (p.3)



1. I can **describe** what pneumatics is and **give** uses of it.
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5. I can **follow** the safety rules for the pneumatics kit.

Extension

1. **Explain** why a pneumatic system needs a **regulator**.

2. **Explain** why port 3 of a kit valve has a **silencer**.

Course Specification — 1 of 1

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 2)

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The big question (a bus door release): Photo: ZidaneHartono, CC BY-SA 4.0, via Wikimedia Commons

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Where Pneumatics Is Used (assembly machine): Photo: Wikimachinespeciale, CC BY-SA 3.0, via Wikimedia Commons

Where Pneumatics Is Used (air impact wrench): Fumikas Sagisavas, CC0, via Wikimedia Commons

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Exam: 2026 Section 1 Question 1: © Qualifications Scotland (SQA). 2026 National 5 Engineering Science Section 1, Q1

Marking: 2026 S1 Q1: © Qualifications Scotland (SQA). 2026 National 5 Engineering Science Section 1, Q1 marking instructions

Credits (2 of 2)

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

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R Stewart - November 2026